1 Method

1.1 Data Collection

1.1.1 Trait selection

We selected traits in the context of three hypotheses. For the predation satiation hypothesis we selected traits associated with seed attractiveness to predators and the potential for surviving in seed caches: seed weight, fruit size, oil content, dispersal mode and seed dormancy. For the pollination coupling hypothesis we aimed for traits related to higher pollination efficiency: pollination mode and reproductive type. For resource matching hypothesis we included traits responsible for higher flexibility in resource use under varying environmental conditions.

1.1.2 Data scraping

For our analyses, we obtained the full species list from the two volumes of *Silvics of North America* Burns & Honkala (1990a,b) (hereafter referred as *Silvics*). We extracted masting data primarily from *Silvics*, using MASTREE as a supplementary source when there was no description on the seed production patterns in *Silvics*. Pollination mode and seed dispersal data were also obtained mainly from *Silvics*. We collected seed weight data from the Seed Information Database by the Society for Ecological Restoration; gathered seed dormancy data from *Seeds* by Baskin and Baskin and accessed the leaf longevity and oil content data through the TRY database. We compiled fruit size and drought tolerance data from multiple sources, including: *Silvics*, the USDA Fire Effects Information System, the USA National Phenology Network, and The Gymnosperm Database.

We defined masting based on relatively qualitative description of seed production patterns. We were aware that recent literature emphasized the importance of quantatative definition of masting, but we are missing such data on the species level. If the description said the species "have a good crop almost every year", we considered this species a weak masting species, whereas if the description said the species " have a good crop every several years or have irregular seed production", we considered this species as a strong masting species.

For fruit size data, we recorded the full range from the source, and took the mean value for further analyses. For pollination mode, we categorized it into wind, animal (bird, bat, or insect), or both. For seed dispersal mode, we categorized it into biotic (animal-dispersed), abiotic (wind-, gravity-, or water-dispersed), or both. For drought tolerance, we categorized it into low, moderate, or high. For leave longeivity, we standardized it to years.

1.2 Statistical Analysis

1.2.1 Masting pattern and single trait

We used `phyloglm` function in the `phylolm` package (v. 2.6.2) to analyze the relationship between masting pattern and individual traits, treating masting as a binary response variable. To construct the phylogenetic tree, we verified all species names using the `WorldFlora` package (v. 1.14-5) and updated all species names according to WCVP Backbone data (2025). We then constructed an ultrametric phylogenetic tree in R using the `phytools` (v. 2.4-4) and `ape` (v. 5.7-1) packages. We calculated the half-life using the equation $(\log(2)/\alpha)$ and compared it with the tree height (standardized to 1). This metric represents the time required for the influence of ancestral states to decrease by half.

1.2.2 Masting pattern with a multivariate analysis

Since most traits in our dataset were categorical, we selected non-metric multidimensional scaling (NMDS) for a multivariate analysis. We used the `metaMDS` function in the `vegan` (v. 2.7-1) package and tested both $k = 2$ and $k = 3$ and selected $k = 3$ based on the lower stress score. We then plotted trait distributions on two panels (NMDS1 vs. NMDS2 and NMDS2 vs. NMDS3) to check for potential patterns.

1.2.3 Phylogenetic signal for individual traits

We calculated phylogenetic signal separately for each trait type. For binary traits, we calculated Fritz and Purvis' D-statistics using the function `phylo.d` in the `caper` (v. 1.0.4) package. For continuous traits, we calculated the Pagel's λ using the function `phylosig` in the `phytools` (v. 2.4-4) package. For categorical traits, we estimated the Pagel's λ using the function `fitDiscrete` in the `geiger` (v. 2.0.11) package.

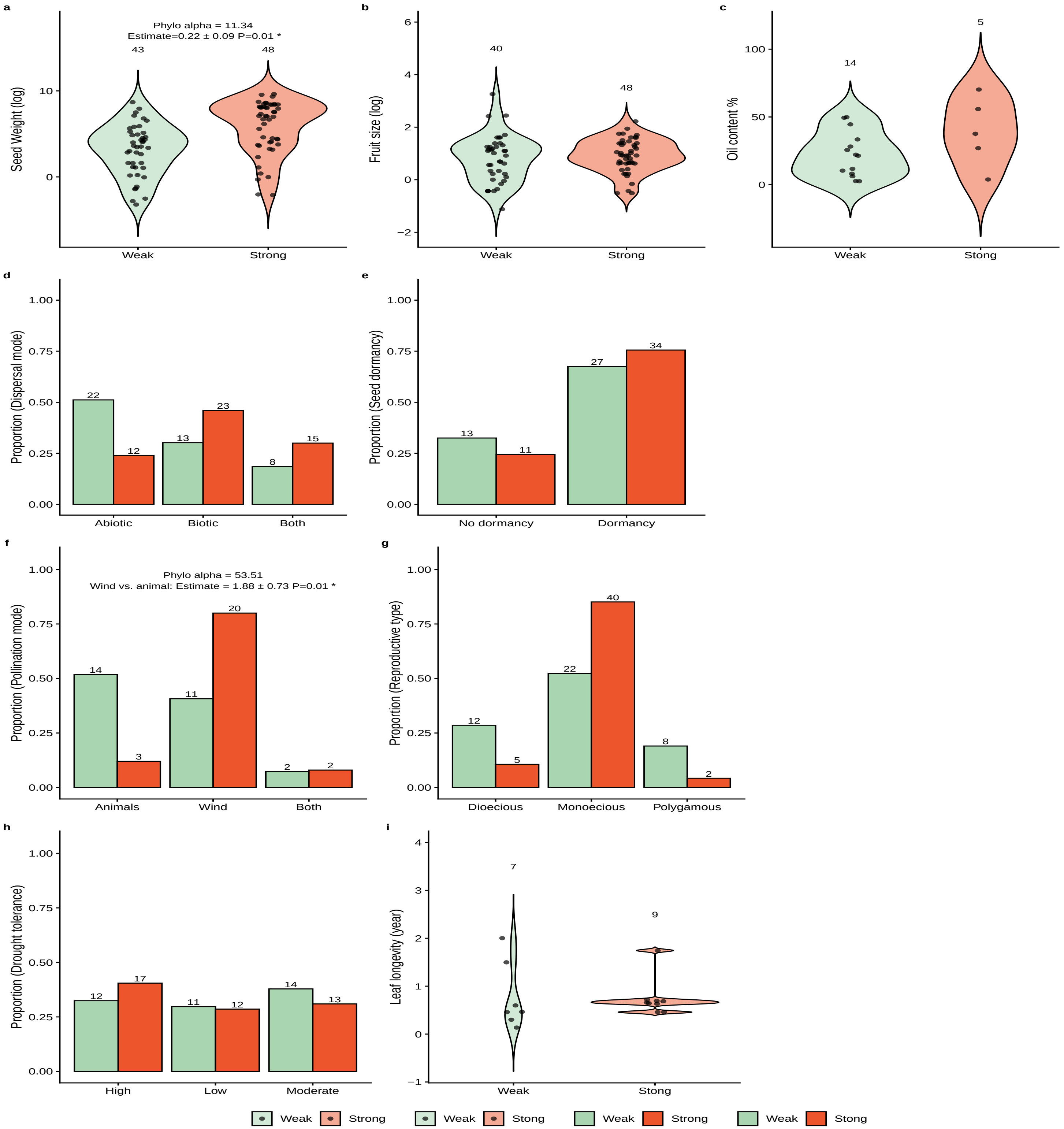

Figure 1: How plant traits related to different hypotheses differ in strong masting and weak masting for angiosperm species: We show data for traits related to the predator satiation hypothesis (a-e), pollination coupling hypothesis (f-g) and the resource matching hypothesis (h-i) as either violin plots (a-c, i) or proportional histograms (d-h) separated by species categorized as strong masting (red) or weak masting (green). We show model summaries only where significant (P-value < 0.5) and also report the model sample size (N), regression coefficient (Estimate), and phylogenetic signal parameter α. Full model results are show in Table S1.

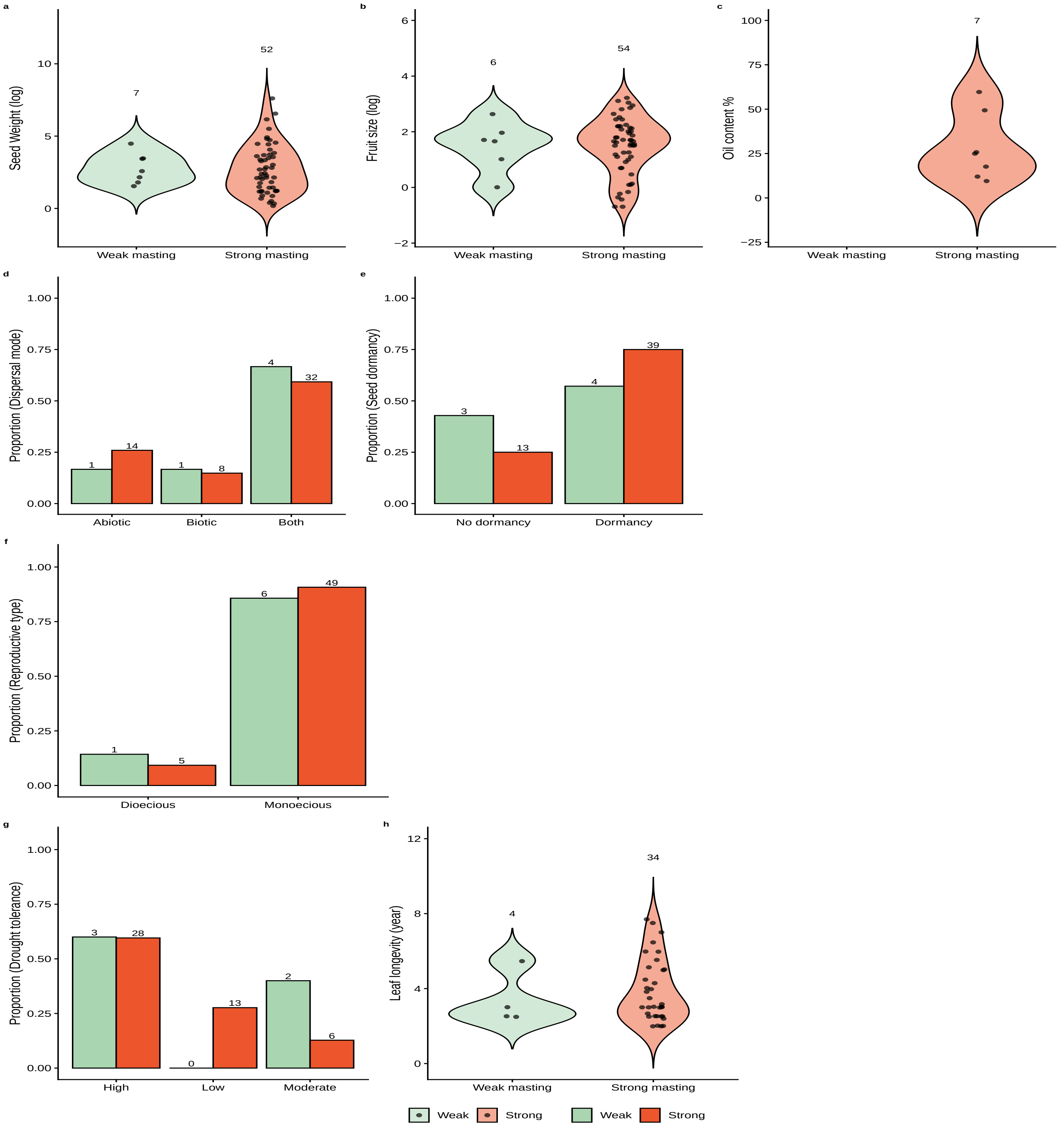

Figure 2: How plant traits related to different hypotheses differ in strong masting and weak masting for gymnosperm species: We show data for traits related to the predator satiation hypothesis (a-e), pollination coupling hypothesis (f-g) and the resource matching hypothesis (h-i) as either violin plots (a-c, i) or proportional histograms (d-h) separated by species categorized as strong masting (red) or weak masting (green). We show model summaries only where significant (P-value < 0.5) and also report the model sample size (N), regression coefficient (Estimate), and phylogenetic signal parameter α. Full model results are show in Table S2.

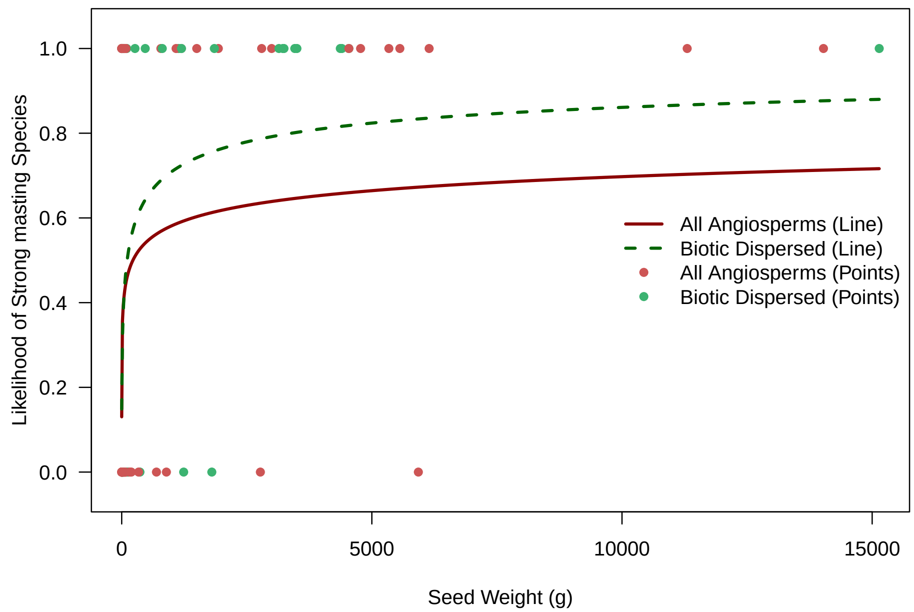

Figure 3: Predicted likelihood of masting as a function of seed weight for all angiosperm species (solid red line) and the biotic-dispersed subgroup (dashed green line), estimated using phylogenetic logistic regression. Individual species observations are plotted as binary data points at 0 (weak masting) and 1 (strong masting).

2 Results

2.0.1 Structure of Data

In total, our dataset included 188 species, comprising 124 angiosperm species and 64 gymnosperm species. Among angiosperms, 79% (98/124) of species had masting data available, with masting data, with strong masting and weak masting split almost equally 52% (51/98). Among gymnosperms, 95% (61/64) of species had masting data available, with almost all species are strong masting 89% (54/61). We were not able to get complete trait data for every single trait for all species with masting data. The most limited data is oil content and leave longevity (Fig. 1 and Fig. 2).

2.0.2 Results for Each Hypothesis

We found relationship between traits related to predator satiation and pollination efficiency and strong masting for angiosperm species. Among traits related to predator satiation, we found seed weight the only significant trait (p-value = 0.01, Table S1) related to masting with larger seeds increasing the probability of being a strong masting species (Figure 1). For angiosperms only, tripling of seed weight (from a median seed weight of 90.9g to 272.7g) is predicted to increase the likelihood of strong masting from 0.45 to 0.51 (Figure 3). And this increase has a larger log-odds in biotic-dispersed seed group, as tripling of seed weight (from a median seed weight of 1087.15g to 3261.45g) is predicted to increase the likelihood of strong masting from 0.72 to 0.80 (Table S3; Figure 3).

Among traits related to pollination efficiency, we found pollination mode the only significant trait (p-value = 0.01, Table S1) related to masting for angiosperm with likelihood for wind-pollinated species is predicted to increase the likelihood of strong masting to 0.62, whereas the likelihood is 0.20 for animal-pollinated species. For gymnosperm, all the species are wind pollinated, and gymnosperm has a greater portion of strong masting species overall.

2.0.3 Multivariate analysis

Masting is not aligned well along any axis in the NMDS analysis. We do not find any evidence that the combination of traits for any of the hypotheses is driving the pattern of masting.

2.0.4 Phylogenetic signal

Strong masting has a moderate phylogenetic signal in both gymnosperms group (estimated D = 0.51) and angiosperms (estimated D = 0.58) group (Table S5). Seed weight has strong phylogenetic signal for both angiosperm (λ = 0.95) and gymnosperm (λ = 0.97). We can see the similarity in certain group in Figure **??**. Dispersal mode, leaf longevity, pollination mode and reproductive type also showed high phylogenetic signal in Table S4, but be cautious that λ for categorical traits is calculated differently and we should not compare them directly with the λ for continuous traits.

References

Burns, R.M. & Honkala, B.H. (eds.) (1990a) *Silvics of North America*, vol. 1 of *Agriculture Handbook*. United States Department of Agriculture, Forest Service, Washington, DC.

Burns, R.M. & Honkala, B.H. (eds.) (1990b) *Silvics of North America*, vol. 2 of *Agriculture Handbook*. United States Department of Agriculture, Forest Service, Washington, DC.

A Supplementary Material

Trait	Type	Predictor	N	Estimate	SE	P	Phylo α	Half-life
Seed weight (log)	Numeric	Seed weight (log)	89	0.22	0.09	0.01	11.34	0.06
Fruit size (log)	Numeric	Fruit size (log)	83	0.00	0.31	0.99	28.28	0.02
Oil content (%)	Numeric	Oil content (%)	19	0.04	0.03	0.20	19.22	0.04
Seed dormancy	Categorical	Dormant vs. non-dormant	81	0.28	0.53	0.60	11.48	0.06
Dispersal mode	Categorical	Biotic vs. abiotic	88	0.69	0.54	0.20	9.86	0.07
Dispersal mode	Categorical	Both vs. abiotic	88	0.66	0.58	0.25	9.86	0.07
Pollination mode	Categorical	Wind vs. animal	48	1.88	0.73	0.01	53.51	0.01
Pollination mode	Categorical	Wind vs. animal and animals	48	-0.25	2.76	0.93	53.51	0.01
Reproductive type	Categorical	Monoecious vs. dioecious	84	0.93	0.69	0.18	13.81	0.05
Reproductive type	Categorical	Monoecious vs. polygamous	84	-0.13	0.96	0.89	13.81	0.05
Drought tolerance	Categorical	Low vs. high drought tolerance	76	0.02	0.46	0.97	8.77	0.08
Drought tolerance	Categorical	Moderate vs. high drought tolerance	76	-0.31	0.46	0.50	8.77	0.08
Leaf longevity (years)	Numeric	Leaf longevity (years)	16	0.26	0.95	0.78	0.90	0.77

Table S1: Phylogenetic generalized linear model results for angiosperm species. We modeled each trait separately with masting as a binary variable and traits as either categorical (Type column), where we report (Estimate column) the the difference in log-odds between a given level and its designated baseline reference level, or continuous, where we report the change in log-odds per unit increase of the predictor (Estimate column). We also report the standard error (SE), P-value (with $P < 0.05$ considered significant and marked with an asterisk next to the Predictor), Phylo α, which represents the the phylogenetic signal in the residuals estimated for each trait, it has an upper bound of 54. If α reached 54, then the phylogenetic correlation is estimated to be negligible. We also calculated half-life ($\log(2)/\alpha$), which represents the time required for the influence of ancestral states to decrease by half, where values greater than 1 also indicate weak phylogenetic signal in the residuals.

Trait	Type	Predictor	N	Estimate	SE	P	Phylo α	Half-life
Seed weight (log)	Numeric	Seed weight (log)	58	-0.01	0.24	0.98	53.48	0.01
Fruit size (log)	Numeric	Fruit size (log)	59	0.03	0.35	0.94	51.67	0.01
Seed dispersal	Categorical	Biotic vs. abiotic	59	-0.82	0.81	0.31	1.99	0.35
Seed dispersal	Categorical	Both vs. abiotic	59	-1.11	0.67	0.10	1.99	0.35
Seed dormancy	Categorical	Dormant vs. non-dormant	58	0.69	0.70	0.33	52.49	0.01
Reproductive type	Categorical	Monoecious vs. dioecious	60	-1.15	1.02	0.26	0.91	0.76
Drought tolerance	Categorical	Low vs. high drought tolerance	51	12.96	249.43	0.96	0.03	23.10
Drought tolerance	Categorical	Moderate vs. high drought tolerance	51	-0.19	0.57	0.74	0.03	23.10
Leaf longevity (years)	Numeric	Leaf longevity (years)	37	0.14	0.34	0.67	54.36	0.01

Table S2: Phylogenetic generalized linear model results for gymnosperm species. We modeled each trait separately with masting as a binary variable and traits as either categorical (Type column), where we report (Estimate column) the the difference in log-odds between a given level and its designated baseline reference level, or continuous, where we report the change in log-odds per unit increase of the predictor (Estimate column). We also report the standard error (SE), P-value (with $P < 0.05$ considered significant and marked with an asterisk next to the Predictor), Phylo α, which represents the the phylogenetic signal in the residuals estimated for each trait, it has an upper bound of 54. If α reached 54, then the phylogenetic correlation is estimated to be negligible. We also calculated half-life ($\log(2)/\alpha$), which represents the time required for the influence of ancestral states to decrease by half, where values greater than 1 also indicate weak phylogenetic signal in the residuals.

Trait	Group	Predictor	N	Estimate	SE	P	Phylo α	Half-life
Seed weight (log)	Biotic and both	Seed weight (log)	53	0.40	0.15	0.01	54.57	0.01
Seed weight (log)	Abiotic	Seed weight (log)	36	0.12	0.13	0.36	50.33	0.01

Table S3: Phylogenetic generalized linear model results for only seed weight for biotic-dispersed subgroup (both group included) and abiotic-dsipersed subgroup. We report the change in log-odds per unit increase of the predictor (Estimate column). We also report the standard error (SE), P-value (with $P < 0.05$ considered significant and marked with an asterisk next to the Predictor), Phylo α, which represents the the phylogenetic signal in the residuals estimated for each trait, it has an upper bound of 54. If α reached 54, then the phylogenetic correlation is estimated to be negligible. We also calculated half-life ($\log(2)/\alpha$), which represents the time required for the influence of ancestral states to decrease by half, where values greater than 1 also indicate weak phylogenetic signal in the residuals.

Group	Trait	λ
Angiosperm	Seed weight (Log)	0.95
Angiosperm	Fruit size (Log)	0.93
Angiosperm	Oil content	0.25
Angiosperm	Leaf longevity	1.02
Angiosperm	Dispersal mode	0.00
Angiosperm	Pollination mode	1.00
Angiosperm	Reproductive type	1.00
Angiosperm	Drought tolerance	0.00
Gymnosperm	Seed weight (Log)	0.97
Gymnosperm	Fruit size (Log)	0.90
Gymnosperm	Oil content %	0.00
Gymnosperm	Leaf longevity	0.90
Gymnosperm	Dispersal mode	0.00
Gymnosperm	Drought tolerance	1.00

Table S4: Phylogenetic Signal (Pagel's λ) for continous traits and multi-level categorical traits. $\lambda = 0$ indicates no phylogenetic signal, trait districuted independent of the phylogeny; $0 < \lambda < 1$ means moderate phylogenetic signal, trait is partially structured by the phylogeny; $\lambda = 1$ indicates strong phylogenetic signal, trait follows Brownian motion.

Group	Trait	Estimated D	P(random)	P(Brownian)
Angiosperm	Seed dormancy	0.25	0	0.16
Angiosperm	Masting	0.58	0	0.01
Gymnosperm	Masting	0.51	0.02	0.10
Gymnosperm	Seed dormancy	0.98	0.41	0
Gymnosperm	Reproductive type	-0.52	0	0.89

Table S5: Phylogenetic Signal (D-statistic) for the phylogenetic signal of binary traits, with probabilities calculated under a random phylogenetic structure and under a Brownian motion phylogenetic structure. $D = 1$ indicates the trait is randomly distributed across the phylogeny; $D = 0$ indicates the trait follows a Brownian motion model; $0 < D < 1$ indivates moderate phylogenetic signal; $D < 0$ indicates the trait is more clumped than expected under Brownian motion; and $D > 1$ indicates overdispersion. P random is the p-value for the null hypothesis that the trait is randomly distributed across the phylogeny (no phylogenetic signal), and P Brownian is the p-value for the null hypothesis that the trait follows a Brownian motion model (strong phylogenetic signal). A low p-value indicates rejection of the corresponding null hypothesis.